

Asymptotic statistics on stratified spaces

Lecture 3: Asymptotic statistics on stratified spaces

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Recap

(S, d) a complete separable metric space; $X_1, X_2, \dots \stackrel{iid}{\sim} P$

- ▶ *Fréchet function*: $F(p) = \int_S d^2(p, x) dP(x)$, $p \in S$.
- ▶ *Fréchet mean set*: $\mu(P) = \arg \min_{p \in S} F(p)$
- ▶ *Sample Fréchet function*: $F_n(p) = \frac{1}{n} \sum_i d^2(p, X_i)$.
- ▶ *Sample Fréchet mean set*: $\mu(P_n) = \arg \min_{p \in S} F_n(p)$

When these are singletons, write $\mu := \mu(P)$ and $\hat{\mu}_n = \mu(P_n)$.

Assume $F(q) < \infty$ for some (hence every) $q \in S$.

Recap

- ▶ Consider unique $\mu, \hat{\mu}_n$
- ▶ Chart $\phi : G \rightarrow U \subseteq \mathbb{R}^s$ at μ for $G \subseteq S$
- ▶ $\hat{\theta}_n = \phi(\hat{\mu}_n)$ is an M-estimator for $m_\theta(x) = d^2(\phi^{-1}(\theta), x)$
- ▶ Conditions for which

$$\sqrt{n}(\phi(\hat{\mu}_n) - \phi(\mu)) \xrightarrow{\mathcal{D}} N_s(0, \Lambda^{-1}\Sigma\Lambda^{-1})$$

(“secret” (A7): compact $\text{supp}(P)$ (??))

- ▶ Manifolds, $\phi = \text{Log}_\mu$
 - ▶ support condition at the cut locus
 - ▶ $\nabla^2 H(\theta_0)$ singular

Today: S not a manifold.

Tools: equivalence relations and quotient spaces

An *equivalence relation* $\sim$ on a set A is a relation that is

- ▶ reflexive: $a \sim a$;
- ▶ symmetric: $a \sim b \implies b \sim a$;
- ▶ transitive: $a \sim b$ and $b \sim c \implies a \sim c$.

The *equivalence class* of a is $[a] = \{b \in A : b \sim a\}$. Equivalence classes partition A .

A *quotient space* $A/\sim$ is the set of equivalence classes, with the map $\pi : A \rightarrow A/\sim$, $\pi(a) = [a]$.

Tools: quotient spaces (ctd)

If A is a topological space, $A/\sim$ carries the *quotient topology*:
 $U \subseteq A/\sim$ is open iff $\pi^{-1}(U)$ is open in A .

If (S, d) is a metric space, $S/\sim$ carries the *quotient (pseudo)metric*

$$\bar{d}([a], [b]) = \inf \left\{ \sum_{i=1}^k d(a_i, b_i) : a_1 \in [a], b_k \in [b], b_i \sim a_{i+1} \right\}.$$

Example 1: the K -spider

Definition: The K -spider is $\mathcal{S}_K = ([0, \infty) \times \{1, \dots, K\}) / \sim$, with $(0, k) \sim (0, j)$ for all j, k

$$d((x, k), (y, j)) = \begin{cases} |x - y| & k = j \\ x + y & k \neq j. \end{cases}$$

Example 1: the K -spider

Consider $\mathcal{S}_3$ and $P_n = \frac{1}{3}(\delta_{(r,1)} + \delta_{(r,2)} + \delta_{(s,3)})$. Then $\mu(P_n) = \{0\}$ for every $s \in [0, 2r]$.

Example 2: open books

Definition: (Hotz et al. 2013) Let $\bar{H}^+ = [0, \infty) \times \mathbb{R}^d$ and $K \geq 3$.
The *open book* is

$$\mathcal{O} = (\bar{H}^+ \times \{1, \dots, K\}) / \sim$$

$$(x, k) \sim (y, j) \iff x = y \text{ and } (k = j \text{ or } x^{(0)} = 0).$$

- ▶ *spine* $L_0 = \{0\} \times \mathbb{R}^d$
- ▶ *leaves* $L_k = \bar{H}^+ \times \{k\}$
- ▶ $R(x^{(0)}, x^{(1)}, \dots, x^{(d)}) = (-x^{(0)}, x^{(1)}, \dots, x^{(d)})$
- ▶ Distance

$$d((x, k), (y, j)) = \begin{cases} \|x - y\|_2 & k = j \\ \|x - Ry\|_2 & k \neq j \end{cases}$$

Example 2: open books

$$d = 1, K = 5$$

Example 3: BHV tree space

$\mathcal{T}_{m+2}$ (Billera–Holmes–Vogtmann 2001): rooted trees (not necessarily binary) with $m + 2$ labelled leaves and positive internal edge lengths.

Example 3: BHV tree space

- ▶ There are $(2m + 1)!!$ possible binary tree topologies
- ▶ For a fixed binary topology, a tree is a point in $\mathbb{R}_{>0}^m$.
- ▶ Sending an edge length to 0 gives a tree with a degree-4 vertex, lying on the boundary of exactly 3 orthants

Example 3: BHV tree space

Informally...

$$\mathcal{T}_{m+2} = \bigsqcup_{\text{topologies}} \mathbb{R}_{\geq 0}^m / \sim$$

Equivalent trees have identical sets of positive-length edges and all positive length edges have the same length.

d = length metric induced by the Euclidean L^2 metric on each orthant

What kind of spaces are these?

Definition: A *stratified space* is a metric space S with a finite partition into nonempty *strata* $S = \bigsqcup_j S_j$ such that:

1. each S_j , in the subspace topology, is a boundaryless manifold of dimension s_j .
2. For $j \neq k$: $\overline{S_j} \cap S_k \neq \emptyset \implies S_k \subseteq \overline{S_j}$ and $s_k < s_j$.

The *codimension* of S_j is $\text{codim } S_j := \max_k s_k - s_j$.

What kind of spaces are these?

The

- ▶ K -spider $\mathcal{S}_K$
- ▶ Open book $\mathcal{O}$ for fixed d and K
- ▶ $\mathcal{T}_{m+2}$

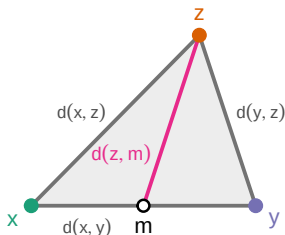
are all stratified spaces.

What can we say about the *geometry* of *these examples*?

Geometry of the examples: nonpositive curvature

Definition: (Sturm 2003) A complete metric space (S, d) is *globally nonpositively curved* (NPC, CAT(0)) if for all $x, y \in S$ there is $m \in S$ such that for every $z \in S$,

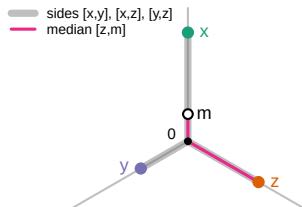
$$d^2(z, m) \leq \frac{1}{2}d^2(z, x) + \frac{1}{2}d^2(z, y) - \frac{1}{4}d^2(x, y).$$



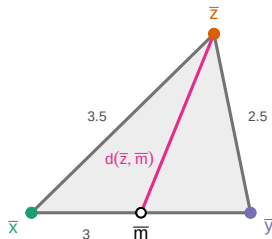
In a NPC space, $d(m, z)$ is no longer than in the Euclidean triangle $\bar{x}\bar{y}\bar{z}$ with the same side lengths (the *comparison triangle*!). Euclidean $m = \frac{x+y}{2}$.

NPC: the 3-spider (“point”, “ray $\in \{1, 2, 3\}$ ”)

3-spider: $d(z, m) = 2$



comparison triangle: $d(\bar{z}, \bar{m}) = \sqrt{7} \approx 2.65$



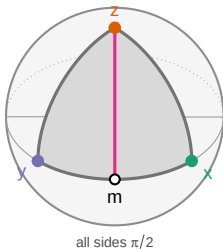
$x = (2, 1)$, $y = (1, 2)$ with midpoint $m = (0.5, 1)$,

Choose $z = (1.5, 3)$:

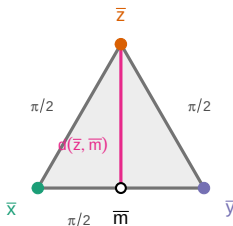
$$d(z, m) = 2 < \sqrt{7} = d(\bar{z}, \bar{m}).$$

Not NPC: the sphere

sphere: $d(z, m) = \pi/2 \approx 1.57$



comparison triangle: $d(\bar{z}, \bar{m}) = \frac{\sqrt{3}}{2} \cdot \frac{\pi}{2} \approx 1.36$



All sides $\pi/2$, and

$$d(z, m) = \pi/2 > \frac{\sqrt{3}}{2} \cdot \frac{\pi}{2} = d(\bar{z}, \bar{m}).$$

S^2 is *not* NPC.

Geodesics and Fréchet means on CAT(0) spaces

Fact: In a CAT(0) space, geodesics are unique!

Theorem: (Sturm 2003) Let S be CAT(0) and $F(q) < \infty$ for some q . **Then F has a unique minimizer μ ,** and for every $p \in S$,

$$F(p) \geq F(\mu) + d^2(p, \mu).$$

Fréchet means for the example spaces

The stratified spaces $\mathcal{S}_K$, $\mathcal{O}$ and $\mathcal{T}_{m+2}$ are all CAT(0), yielding unique sample and population Fréchet means.

From Ziezold's consistency for metric spaces, we have $\hat{\mu}_n \xrightarrow{a.s.} \mu$ in CAT(0) spaces.

We *don't* have unique means for all measures on stratified spaces (e.g. all manifolds are stratified spaces).

Sticky means

$\mathcal{P}_2(S)$: probability measures on S with $F(q) < \infty$. The *2-Wasserstein distance* gives a distance between measures on S :

$$W_2(P, Q) = \inf_{\pi \in \Gamma(P, Q)} \left(\int_{S \times S} d^2(x, y) d\pi(x, y) \right)^{1/2}$$

where $\Gamma(P, Q)$ is the set of joint measures π on $S \times S$ with marginals P and Q .

Definition: Let $S = \bigsqcup_j S_j$ be stratified and $P \in \mathcal{P}_2(S)$ have Fréchet mean $\mu \in S_j$. The mean of P is *sticky on S_j* if there is $\varepsilon > 0$ such that

$$W_2(P, Q) < \varepsilon \implies \mu(Q) \subseteq S_j.$$

Sticky means

Stickiness arises in stratified NPC spaces and has been characterized in important examples.

Open books $\mathcal{O}$: Sticky means

Definition: Folding map $\Phi_k : \mathcal{O} \rightarrow \mathbb{R}^{d+1}$,

$$\Phi_k(x, j) = \begin{cases} x & j = k \\ Rx & j \neq k. \end{cases}$$

Definition: First moment on leaf: With $\pi_0(x) = x^{(0)} \in \mathbb{R}^{d+1}$ (i.e., drop the page label),

$$m_k = \int_{\mathcal{O}} \pi_0 \Phi_k(x) dP(x) = \int_{L_k^+} x^{(0)} dP - \sum_{j \neq k} \int_{L_j^+} x^{(0)} dP.$$

Characterizing mean behaviour

Theorem: (Hotz et al. 2013) Exactly one of the following holds:

1. $m_j < 0$ for all $j \in \{1, \dots, K\}$;
▶ “Sticky”
2. $m_k > 0$ for exactly one k (and $m_j < 0$ for $j \neq k$);
▶ “Non-sticky”
3. $m_k = 0$ for exactly one k (and $m_j < 0$ for $j \neq k$).
▶ “Partly sticky”

Limiting distributions

On stratified NPC spaces, we have unique and consistent Fréchet means.

To talk about distributions, we need a *chart*:

$G \subseteq S$, $\mu \in G$, a homeomorphism $\phi : G \rightarrow U \subseteq \mathbb{R}^s$ for U open.

Then, we'll study $\sqrt{n}(\phi(\hat{\mu}_n) - \phi(\mu))$.

Which chart?

If μ is in a top-dimensional stratum S_j , everything is “easy”!

Choose

- ▶ $G \subseteq S_j$ a neighbourhood of μ
- ▶ $\phi = \text{Log}_\mu$

and use the manifold CLT!

Open books: Sticky CLT

What is $\log_\mu$ on an open book?

- ▶ $\mu \in L_k^+$ (nonsticky): the folding map $\Phi_k : \mathcal{O} \rightarrow \mathbb{R}^{d+1}$, an isometry on $L_k \cup L_j$ for every j .
 - ▶ Study $\sqrt{n}(\Phi_k \hat{\mu}_n - \Phi_k \mu)$.
- ▶ $\mu \in L_0$ (sticky, partly sticky): translation along the spine is an isometry of $\mathcal{O}$, so WLOG $\mu = \mathbf{0}$.
 - ▶ Study $\sqrt{n} \hat{\mu}_n \in \mathcal{O}$

Open books: Sticky CLT

Theorem: (Hotz et al. 2013) Let P be nondegenerate with $F(\mathbf{0}) < \infty$.

1. sticky: $\sqrt{n} \hat{\mu}_n \xrightarrow{\mathcal{D}} N_d(0, C_S)$ on the spine $L_0 \cong \mathbb{R}^d$
 - $C_S = \text{Var}(P_S X)$, where $P_S(x^{(0)}, \tilde{x}, k) = \tilde{x}$ is the projection onto the spine
2. nonsticky, $m_k > 0$: $\sqrt{n}(\Phi_k \hat{\mu}_n - \Phi_k \mu) \xrightarrow{\mathcal{D}} N_{d+1}(0, C_k)$
 - $C_k = \text{Var}(\Phi_k X)$
3. partly sticky, $m_k = 0$: $\sqrt{n} \hat{\mu}_n \xrightarrow{\mathcal{D}} \Phi_k^{-1} \hat{P} Z_k$ on L_k , where $Z_k \sim N_{d+1}(0, C_k)$
 - $\hat{P}(x^{(0)}, \tilde{x}) = (\max\{x^{(0)}, 0\}, \tilde{x})$, the convex projection of $\mathbb{R}^{d+1}$ onto $\bar{H}^+$

Tree space: μ in a top-dimensional stratum

Suppose $\mu \in \mathcal{T}_{m+2}$ is binary with edge-length coordinates $t^* \in \mathbb{R}_{>0}^m$.

The geodesic from μ to x is a straight segment in the orthant of μ , then a sequence of straight segments through neighbouring orthants.

Tree space: μ in a top-dimensional stratum

Write $\log_{\mu} x = d(\mu, x) v(x) \in \mathbb{R}^m$ with $v(x)$ the initial direction, and

$$\ell_{\mu}(x) := t^* + \log_{\mu} x \in \mathbb{R}^m$$

Lemma: (Barden, Le & Owen 2018) μ in a top-dimensional stratum is the Fréchet mean of P iff $\int \log_{\mu} x dP(x) = 0$.

For fixed x , the algebraic form of $\log_{\mu} x$ is locally constant in μ .
Let $D_{\mu} \subseteq \mathcal{T}_{m+2}$ be the set of x at which the form changes.

CLT: top-dimensional stratum

Theorem: (BLO 2018) Let P on $\mathcal{T}_{m+2}$ have $F < \infty$ and Fréchet mean μ in a top-dimensional stratum, with $P(D_\mu) = 0$. Let $\hat{t}_n$ be the coordinates of $\hat{\mu}_n$. Then

$$\sqrt{n} (\hat{t}_n - t^*) \xrightarrow{\mathcal{D}} N_m(0, AVA)$$

where

$$V = \text{Var}(\log_\mu X_1)$$

and

$$A = (I - \mathbb{E} M_\mu(X_1))^{-1}$$

for $M_\mu(x) \in \mathbb{R}^{m \times m}$ the derivative of $\nu \mapsto \ell_\nu(x)$ (ℓ the translated log map).

$P(D_\mu) = 0$ plays the role of the [support condition](#).

Codimension ≥ 1

BLO 2018 also gave the limiting behaviour when μ lies in a codimension-1 stratum, but noted that “a complete description of the limiting behaviour of sample Fréchet means in the wider context is still a challenge.”

Mattingly, Miller and Tran filled this gap in a sequence of preprints released in 2023.

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Three metric spaces

Stratified spaces & NPC spaces

Behaviour of means: consistency and stickiness

Distributional results

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